Chapter 5

MULTISTAGE, POWER, AND DIFFERENTIAL AMPLIFIERS

CHAPTER OUTLINE

5-1 Coupled Amplifiers

5-2 Power Amplifiers

5-3 Differential Amplifiers

5-4 Frequency Response of Amplifiers

5-5 Negative Feedback in Amplifiers

5-4 Frequency Response of Amplifiers

5-4-1 Analysis Methods of Frequency Response

5-4-2 High Frequency Model of Transistors

5-4-3 Frequency Response Analysis of Amplifiers

5-4-4 Methods of Expanding Bandwidth

5-4-5 Brief Summary

refer to chapter 8 of the textbook

5-4-1 Analysis methods of frequency response

- Concepts
- Transfer Function
- Bode Plot

Concepts

Definition of Frequency Response

-- steady-state response of an amplifier to input sinusoidal signal.
-- reflects the ability to amplify signals at different frequencies.
-- denoted as frequency characteristics, $A(j\omega)$ or $A(jf)$

- Frequency characteristics
  -- magnitude frequency characteristics: $|A(j\omega)|$ or $|A(jf)|$
  -- Phase frequency characteristics: $\phi(\omega)$ or $\phi(f)$

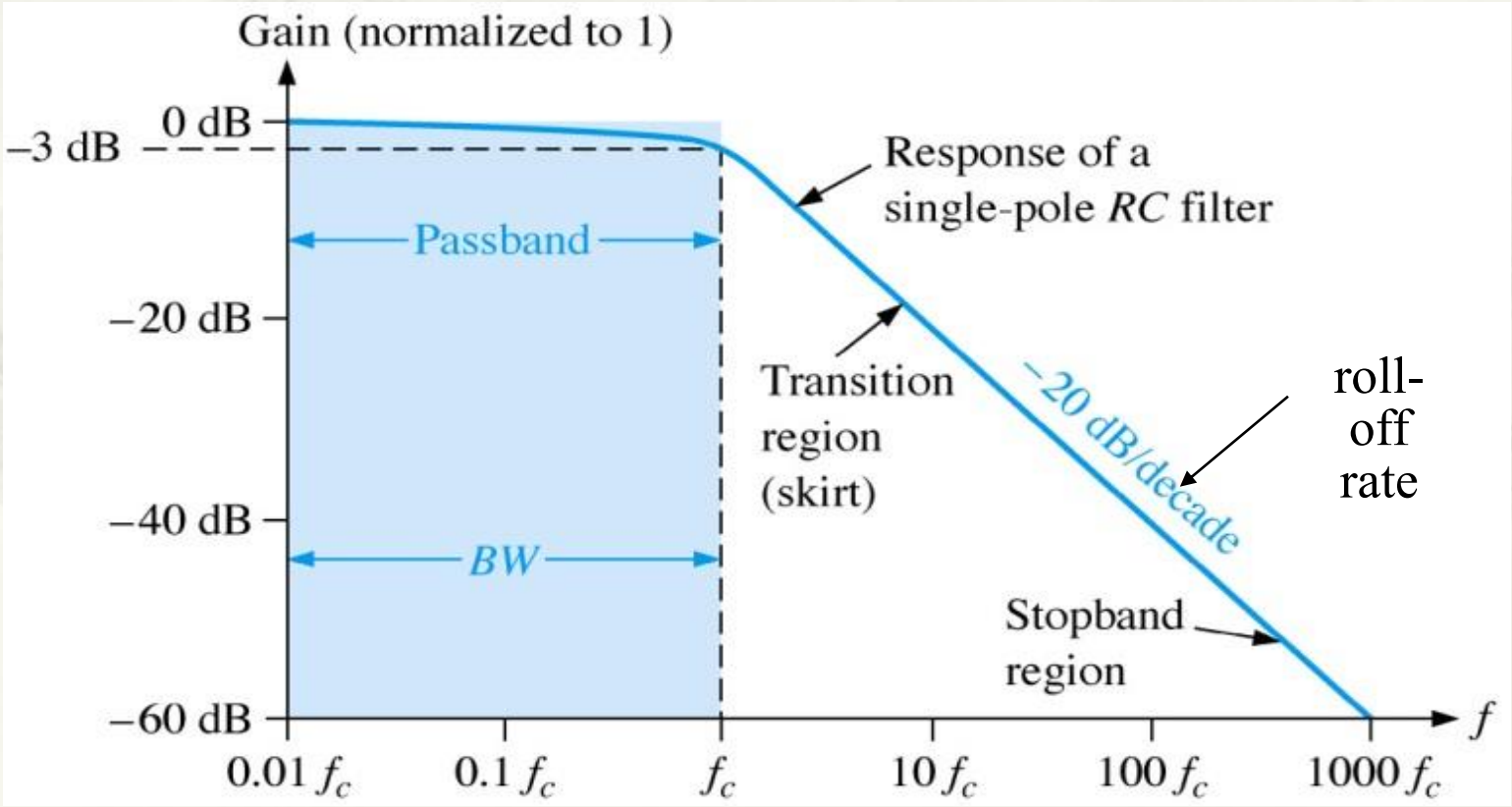
Concepts

Frequency Characteristic Curves

- magnitude frequency characteristic curve

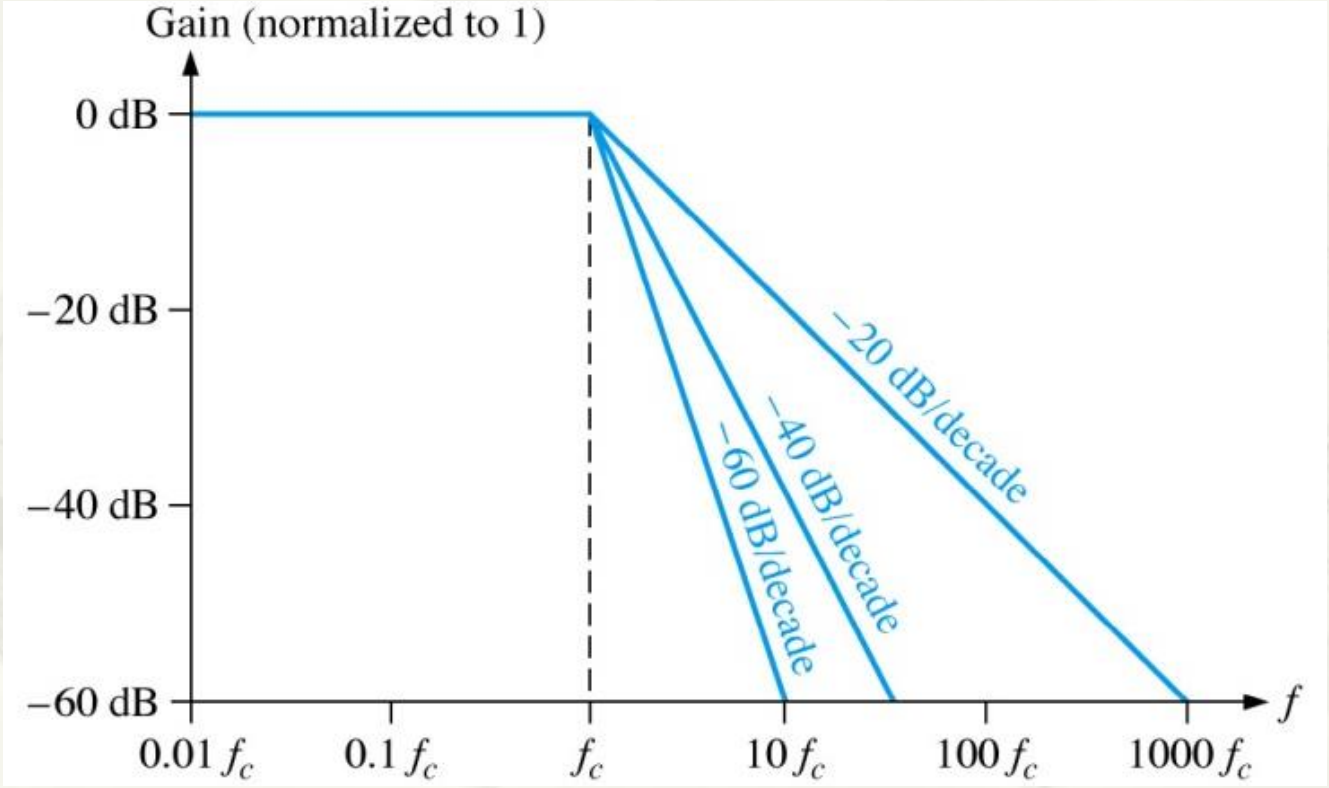
  --describe that the voltage **gain** varies with frequency

  --expressed as $|A(jf)|$ or $|A(j\omega)|$



Concepts

- **Frequency Characteristic Curves**
  - magnitude frequency characteristic curve



different roll-off rates

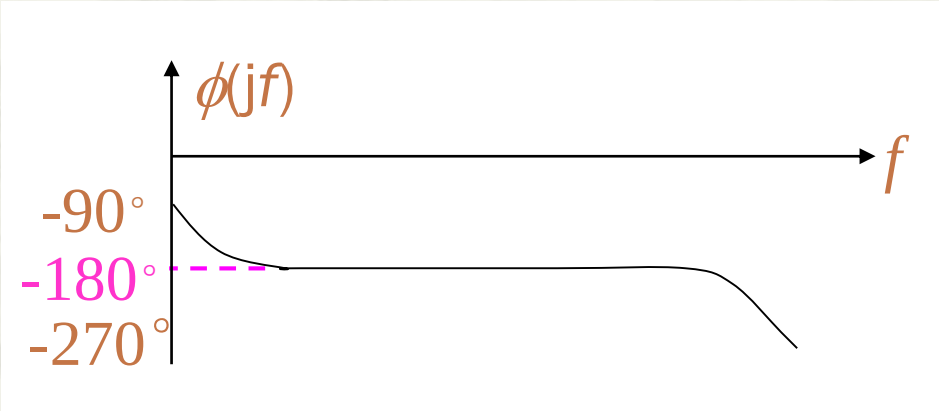
Concepts

Frequency Response Curves

- Phase frequency characteristic curve

  -- describe that the phase difference between the output signal and the input signal varies with frequency

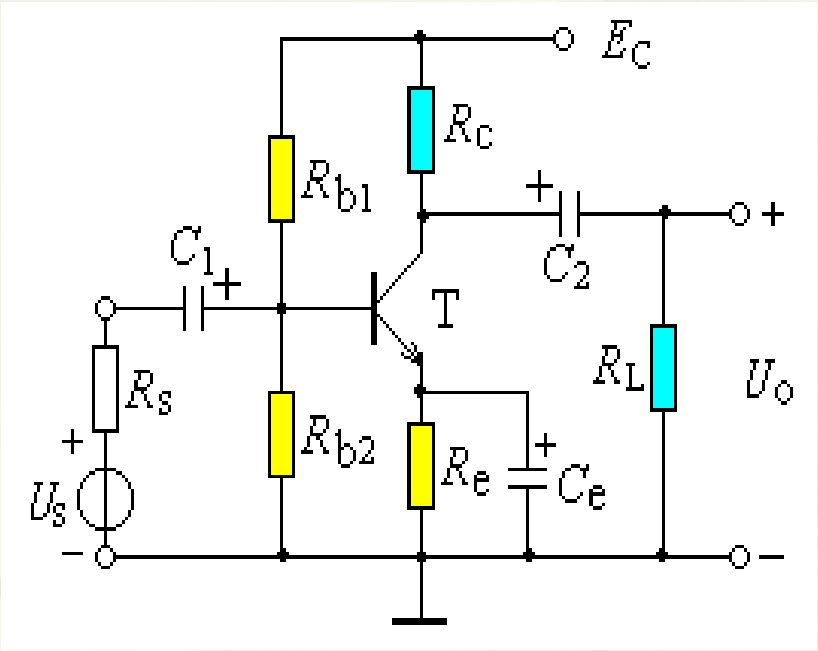
  -- expressed as $|\phi(\omega)|$



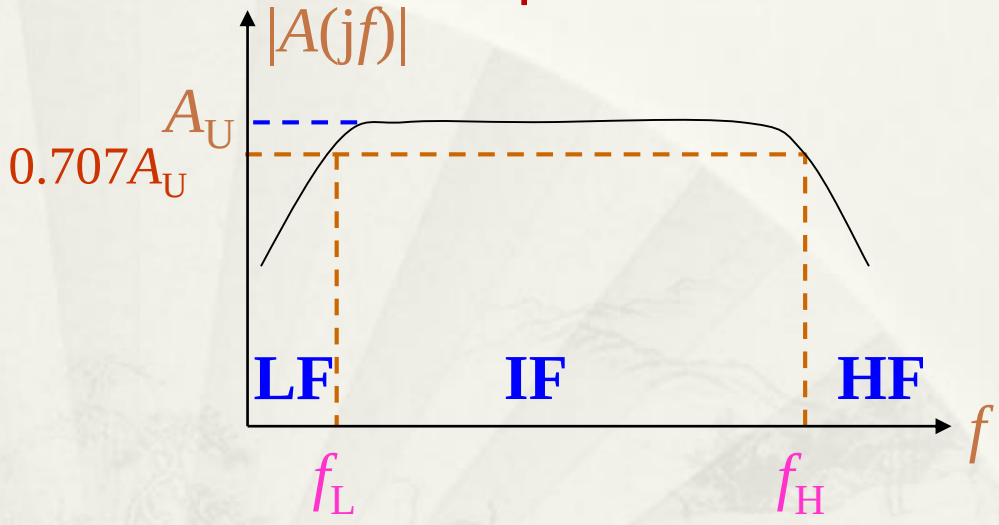
Concepts

Frequency Response Curves

Frequency Response Curves of a CE Amplifier



A CE amplifier





LF -- low frequency
IF -- intermediate frequency
HF -- high frequency

Concepts

Effects on Frequency Characteristics

- reactance components in an amplifier

  -- Such as coupling capacitance, by-pass capacitance, distributed capacitance, transformer, distributed inductance, etc.

  Affect LF gain -- coupling capacitance, by-pass capacitance

  Affect HF gain -- junction capacitance in transistors, and wire capacitance

- $\beta(\omega)$ is the function of frequency

  -- In the study of frequency characteristic, the LF small signal model of a transistor is no longer applicable, but the HF small signal model should be adopted.

Transfer Function

- **Analysis of Linear Systems**



Transfer function

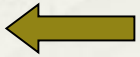
- **Transfer Function of Linear systems**

$$H(s)=\frac{Y(s)}{F(s)}=H_0\frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}$$

$z_1, z_2, \ldots, z_m$ ---- zero

$p_1, p_2, \ldots, p_n$ ---- pole

$$H(s)=H_0'\bullet\frac{(1-\frac{s}{z_1})(1-\frac{s}{z_2})\cdots}{(1-\frac{s}{p_1})(1-\frac{s}{p_2})\cdots}$$

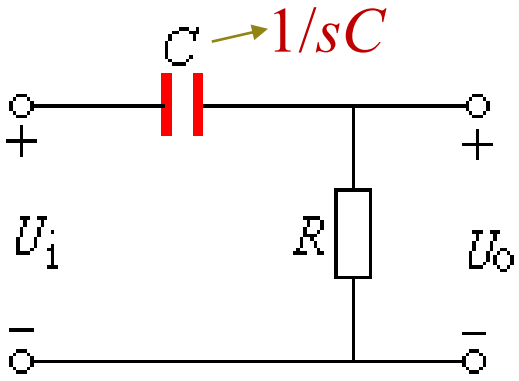


standard expression for bode plot

Transfer function

- **Example**

$$U_o(s) = \frac{R}{R+1/(sC)} U_i(s)$$



RC high-pass circuit

$$H(s) = \frac{U_o(s)}{U_i(s)} = \frac{sRC}{sRC+1} = \frac{s}{s+1/RC} = \frac{s-z_1}{s-p_1}$$

zero: $z_1 = 0$ ploe: $p_1 = -1/RC$

Let $s = j\omega$

$$A_u(j\omega) = \frac{U_o(j\omega)}{U_i(j\omega)} = \frac{j\omega - z_1}{j\omega - p_1}$$

---- frequency characteristics (FC)

$$= \frac{\omega RC}{\sqrt{1+\omega^2R^2C^2}} \angle arctan\frac{-1}{\omega RC}$$

---- magnitude FC

---- phase FC

Bode Plot

- **Definition**

  -- describe frequency response characteristics in a logarithmic coordinate system.

- **Asymptotic line description**

  -- approximately describe frequency response characteristics.

Step 1: rewrite the transfer function

$$A(\mathrm{j}\omega)=\frac{A_1\,\mathrm{j}\omega(\mathrm{j}\omega+\omega_1)}{(\mathrm{j}\omega+\omega_2)(\mathrm{j}\omega+\omega_3)(\mathrm{j}\omega+\omega_4)}$$

$$A(\mathrm{j}\omega)=\frac{A\,\mathrm{j}\omega\left(1+\mathrm{j}\dfrac{\omega}{\omega_1}\right)}{\left(1+\mathrm{j}\dfrac{\omega}{\omega_2}\right)\left(1+\mathrm{j}\dfrac{\omega}{\omega_3}\right)\left(1+\mathrm{j}\dfrac{\omega}{\omega_4}\right)}$$

Bode Plot

- **Asymptotic line description**

Step 1: rewrite the transfer function

Step 2: rewrite the transfer function in logarithm

$$A(\mathrm{j}\omega)=\frac{A\,\mathrm{j}\omega\left(1+\mathrm{j}\frac{\omega}{\omega_1}\right)}{\left(1+\mathrm{j}\frac{\omega}{\omega_2}\right)\left(1+\mathrm{j}\frac{\omega}{\omega_3}\right)\left(1+\mathrm{j}\frac{\omega}{\omega_4}\right)}$$

magnitude frequency characteristic

$$20\lg\left|A(\mathrm{j}\omega)\right|=20\lg A+20\lg\left|\mathrm{j}\omega\right|+20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_1}\right|$$

$$-20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_2}\right|-20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_3}\right|-20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_4}\right|$$

Bode Plot

Asymptotic line description

Step 1: rewrite the transfer function

Step 2: rewrite the transfer function in logarithm

$$A(\mathrm{j}\omega)=\frac{A\,\mathrm{j}\omega\left(1+\mathrm{j}\frac{\omega}{\omega_1}\right)}{\left(1+\mathrm{j}\frac{\omega}{\omega_2}\right)\left(1+\mathrm{j}\frac{\omega}{\omega_3}\right)\left(1+\mathrm{j}\frac{\omega}{\omega_4}\right)}$$



phase frequency characteristic

$$\angle A(j\omega)=\angle A+\angle(j\omega)$$

$$+\arctan\left(\frac{\omega}{\omega_1}\right)-\arctan\left(\frac{\omega}{\omega_2}\right)-\arctan\left(\frac{\omega}{\omega_3}\right)-\arctan\left(\frac{\omega}{\omega_4}\right)$$

Bode Plot

Asymptotic line description

Step 3: plot asymptotic lines for magnitude frequency characteristic

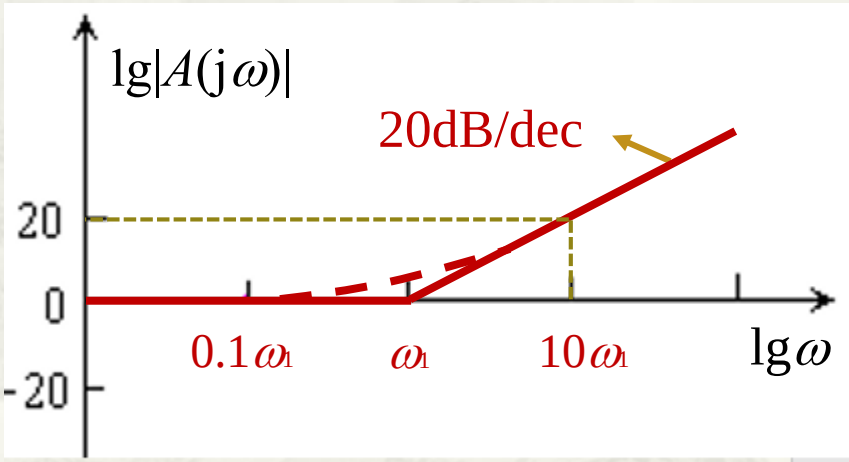
3.1 for the first-order zero $(1+\mathrm{j}\frac{\omega}{\omega_1})$

$$20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_1}\right| = 20\lg\sqrt{1+(\frac{\omega}{\omega_1})^2} = y$$

when $\omega << \omega_1$, $y \approx 20\lg 1 = 0$ dB

when $\omega >> \omega_1$, $y \approx 20\lg(\omega/\omega_1)$

when $\omega = \omega_1$, $y = 3$ dB



Bode Plot

Asymptotic line description

Step 3: plot asymptotic lines for magnitude frequency characteristics

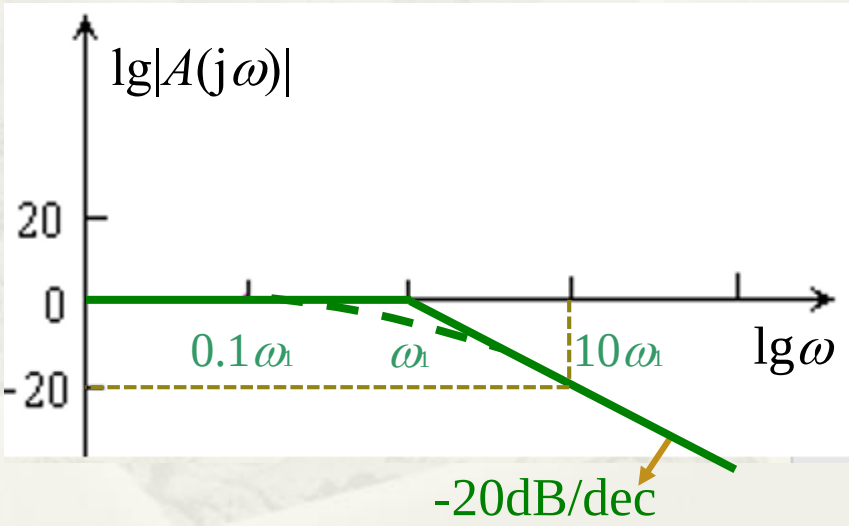
3.2 for the first-order pole $(1+\mathrm{j}\frac{\omega}{\omega_1})$

$$-20\lg\left|1+\mathrm{j}\frac{\omega}{\omega_1}\right| = -20\lg\sqrt{1+(\frac{\omega}{\omega_1})^2} = y$$

when $\omega << \omega_1$, $y \approx -20\lg 1 = 0$ dB

when $\omega >> \omega_1$, $y \approx -20\lg(\omega/\omega_1)$

when $\omega = \omega_1$, $y = -3$ dB



Bode Plot

Asymptotic line description

Step 3: plot asymptotic lines for magnitude frequency characteristics

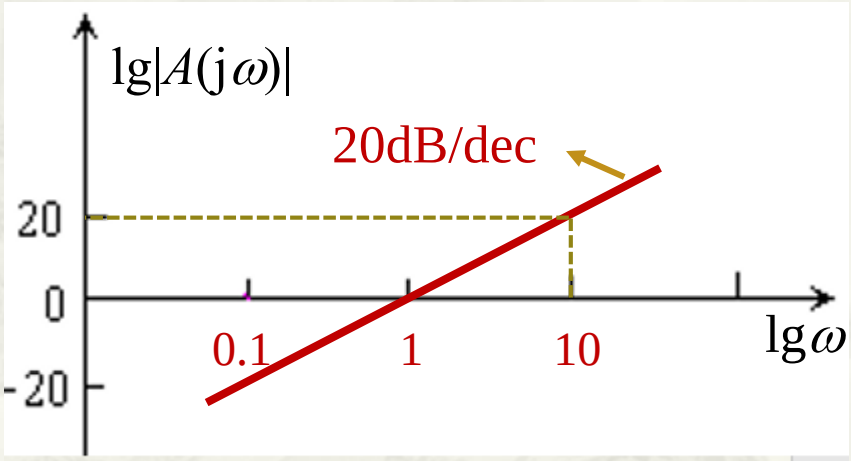
3.3 for the first-order zero ($j\omega$)

$$j\omega = j\frac{\omega}{\omega_0}\Big|_{\omega_o=1}$$

$$20\lg|j\omega| = 20\lg\frac{\omega}{\omega_0} = y$$

when $\omega = \omega_0 = 1$, $y = 0$ dB

when $\omega = 10\omega_0$, $y = 20$ dB



So, it's a straight line passing through the point ($\omega = \omega_0 = 1$), and the slope is 20 dB/decade

Bode Plot

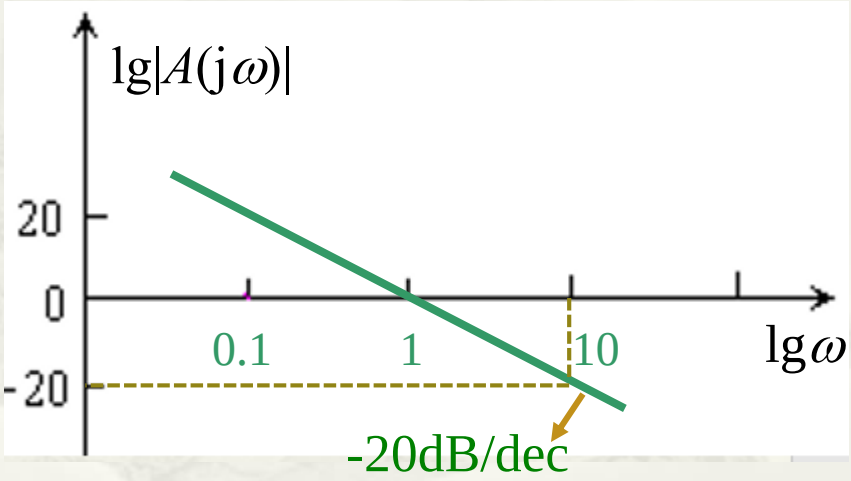
Asymptotic line description

Step 3: plot asymptotic lines for magnitude frequency characteristics

3.4 for the first-order pole $(j\omega)$

$$-20\lg\left|j\omega\right| = -20\lg\frac{\omega}{\omega_0} \quad = y$$

when $\omega = \omega_0 = 1$, $y = 0$ dB
when $\omega = 10\omega_0$, $y = -20$ dB



So, it's a straight line passing through the point ($\omega = \omega_0 = 1$), and the slope is -20 dB/decade

Bode Plot

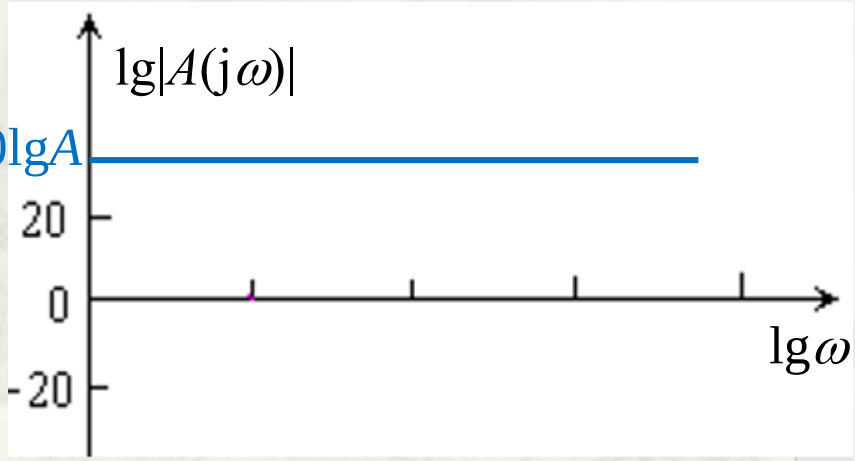
Asymptotic line description

Step 3: plot asymptotic lines for magnitude frequency characteristics

3.5 for the constant term (A)

$20\lg A$ = constant

So, it's a horizontal straight line with the magnitude $20\lg A$



3.6 superimpose all lines

Summary: { the slope increases 20 dB/dec with a zero
the slope decreases 20 dB/dec with a pole

Bode Plot

Asymptotic line description

Example 1:

$$A(\mathrm{j}\omega)=\frac{10^5}{(1+\mathrm{j}\frac{\omega}{10^4})(1+\mathrm{j}\frac{\omega}{10^6})(1+\mathrm{j}\frac{\omega}{10^7})}$$

Solution:

1. It is a standard equation for bode plot
2. $A=10^5 \longrightarrow 20\lg A = 20\lg 10^5 = 100$ dB $\longrightarrow$ a horizontal line
3. 3 poles: 10^4, 10^6 and 10^7 $\longrightarrow$ 3 asymptotic lines
4. superimpose 4 lines together

Bode Plot

Asymptotic line description

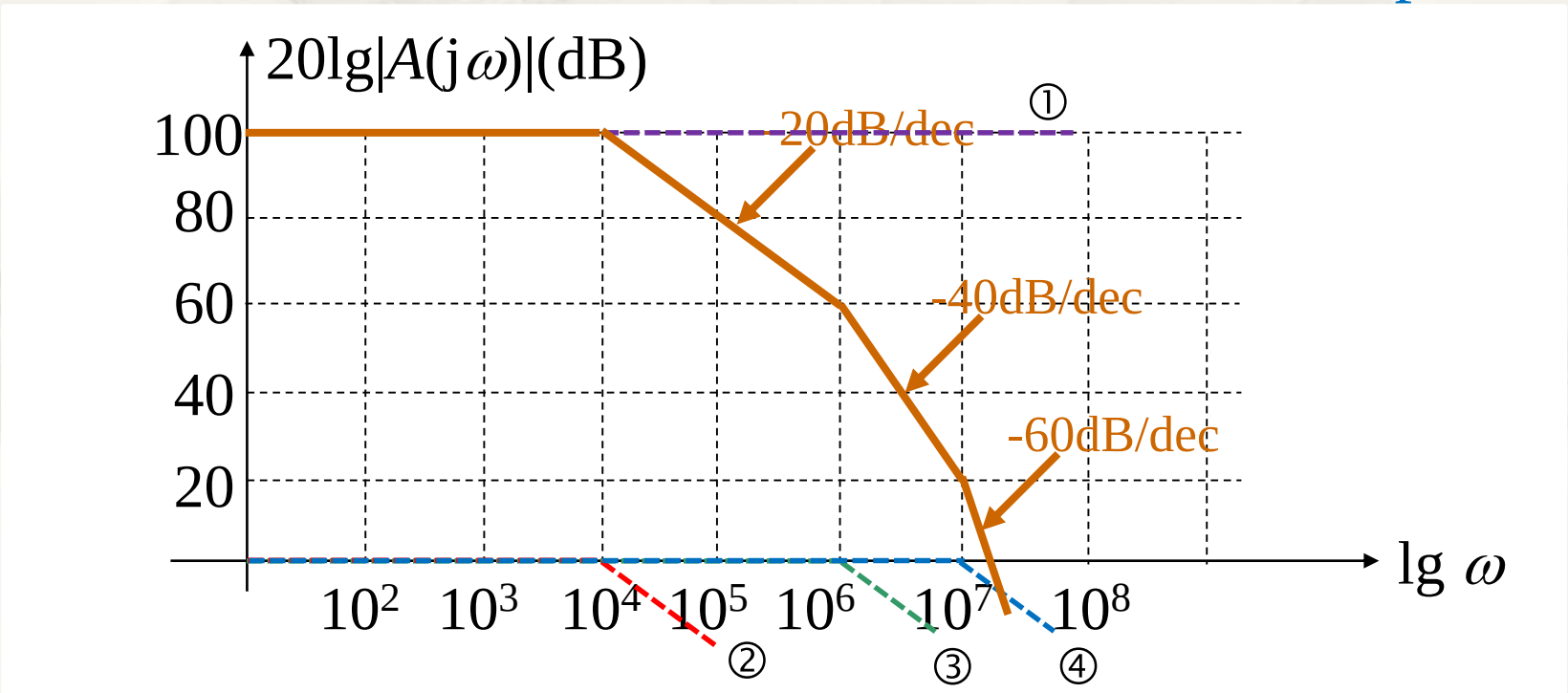
Example 1:

$$A(j\omega)=\frac{10^5}{(1+j\frac{\omega}{10^4})(1+j\frac{\omega}{10^6})(1+j\frac{\omega}{10^7})}$$

① 10^5 → constant 100 dB

② 10^4, ③ 10^6, ④ 10^7 → 3 poles

Solution:



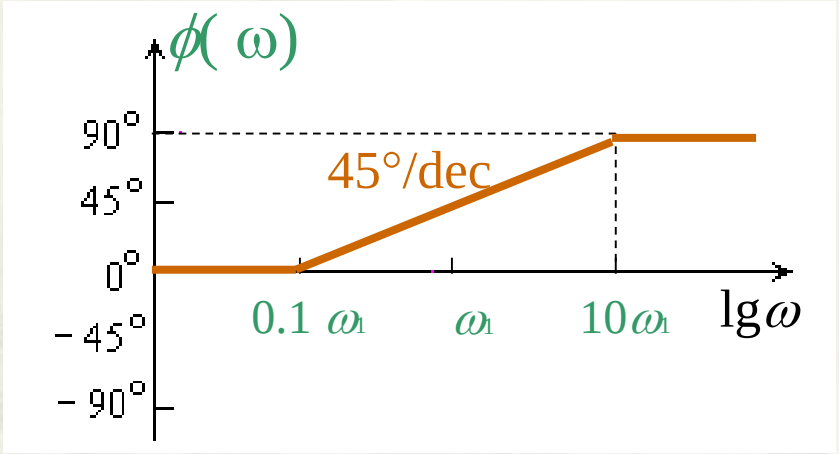
Bode Plot

Asymptotic line description

Step 4: plot asymptotic lines for phase frequency characteristics

4.1 for the first-order zero $(1+\mathrm{j}\frac{\omega}{\omega_1})$

$$\phi(\omega) = \arctan(\frac{\omega}{\omega_1})$$



when $\omega << \omega_1$, $\phi(\omega) = 0$

when $\omega = \omega_1$, $\phi(\omega) = 45^\circ$

when $\omega >> \omega_1$, $\phi(\omega) = 90^\circ$

3 asymptotic lines:

- when $\omega \le 0.1\omega_1$, 0^0 horizontal line
- when $\omega \ge 10\omega_1$, 90^0 horizontal line
- when $0.1\omega_1 \le \omega \le 10\omega_1$, 45^0/dec slash line

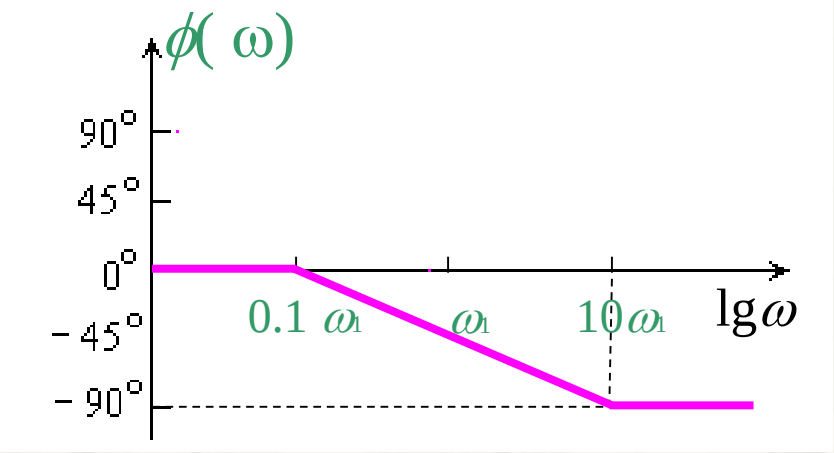
Bode Plot

Asymptotic line description

Step 4: plot asymptotic lines for phase frequency characteristics

4.2 for the first-order pole $(1+\mathrm{j}\frac{\omega}{\omega_1})$

$$\varphi(\omega) = -\arctan(\frac{\omega}{\omega_1})$$



when $\omega << \omega_1$, $\phi(\omega) = 0$

when $\omega = \omega_1$, $\phi(\omega) = -45^\circ$

when $\omega >> \omega_1$, $\phi(\omega) = -90^\circ$

3 asymptotic lines:

- when $\omega \le 0.1\omega_1$, 0^0 horizontal line
- when $\omega \ge 10\omega_1$, -90^0 horizontal line
- when $0.1\omega_1 \le \omega \le 10\omega_1$, -45^0/dec backslash line

Bode Plot

Asymptotic line description

Step 4: plot asymptotic lines for phase frequency characteristics

4.3 for the first-order zero ($j\omega$)

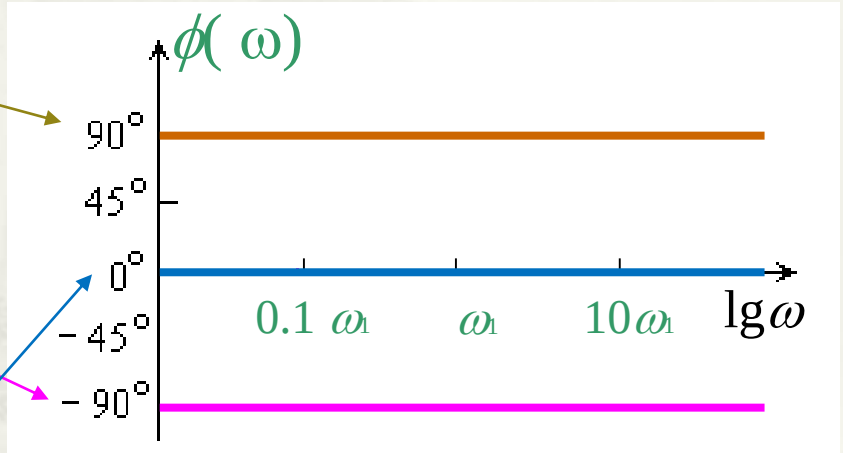
$\varphi(\omega) = 90°$ a 90^0 horizontal line

4.4 for the first-order pole ($j\omega$)

$\varphi(\omega) = -90°$ a -90^0 horizontal line

4.5 for the constant term (A)

$\varphi(\omega) = 0°$ don't need a line



Summary: { for a zero, positive phase shift, the largest one is + 90°
for a pole, negative phase shift , the largest one is - 90°

Bode Plot

Asymptotic line description

Example 1:

$$A(j\omega)=\frac{10^5}{(1+j\frac{\omega}{10^4})(1+j\frac{\omega}{10^6})(1+j\frac{\omega}{10^7})}$$

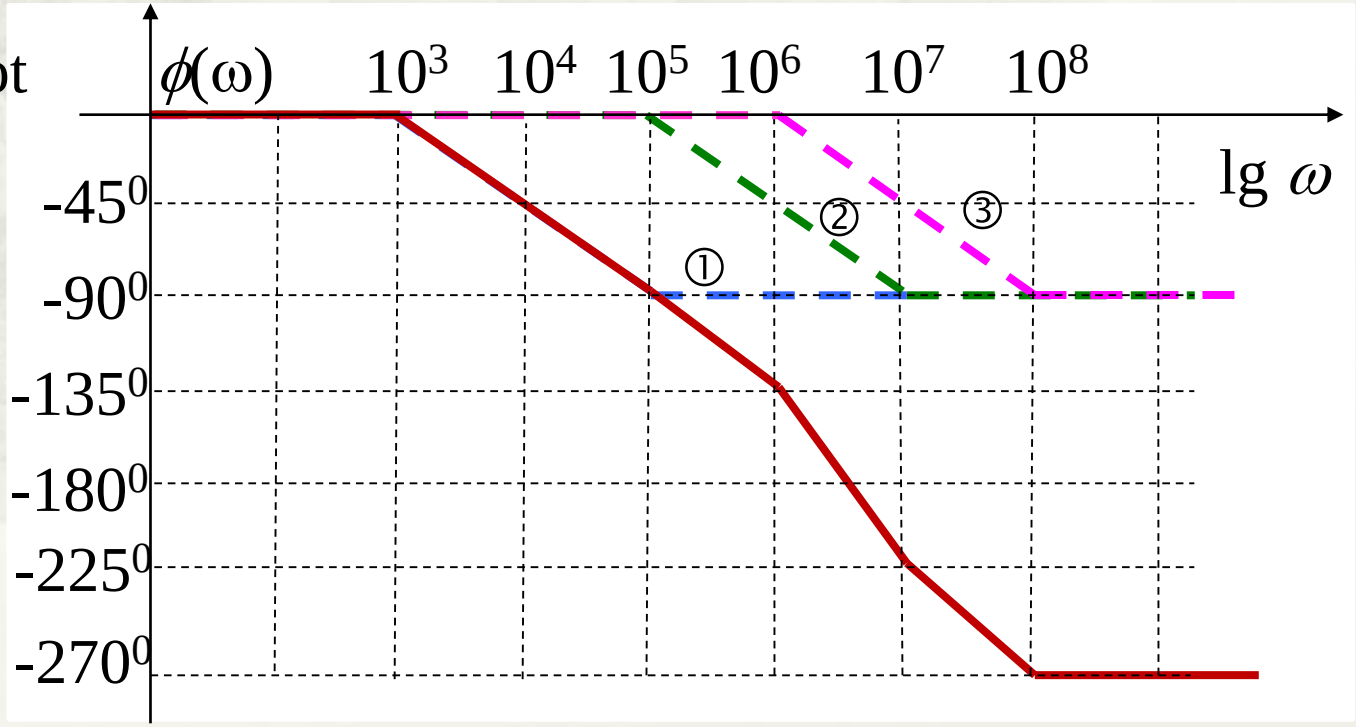
① ② ③

Solution:

1. It is a standard equation for bode plot

2. 3 poles: 10^4, 10^6 and 10^7

3. superimpose 3 lines together



Bode Plot

Asymptotic line description

Example 2:

$$A(\mathrm{j}\omega)=\frac{2\times10^{6}\, j\omega(\mathrm{j}\omega+10)}{(\mathrm{j}\omega+20)(\mathrm{j}\omega+100)(\mathrm{j}\omega+10^{4})}$$

Solution:

1. rewrite the equation

$$A(\mathrm{j}\omega)=\frac{\mathrm{j}\omega(1+\mathrm{j}\frac{\omega}{10})}{(1+\mathrm{j}\frac{\omega}{20})(1+\mathrm{j}\frac{\omega}{100})(1+\mathrm{j}\frac{\omega}{10^{4}})}$$

2. constant term: A=1 ⟶ 20lg1 = 0 dB ⟶ don't need a line

3. 2 zeros: 1, and 10 ⟶ 2 asymptotic lines

4. 3 poles: 20, 10^2 and 10^4 ⟶ 3 asymptotic lines

5. superimpose 6 lines together

Bode Plot

- **Asymptotic line description**

constant 0 dB

Example 2:

$$A(j\omega)=\frac{j\omega\left(1+j\frac{\omega}{10}\right)}{\left(1+j\frac{\omega}{20}\right)\left(1+j\frac{\omega}{100}\right)\left(1+j\frac{\omega}{10^4}\right)}$$

② ① 2 zeros

⑤ ④ ③ 3 poles

Solution:

f_L -- largest LF pole

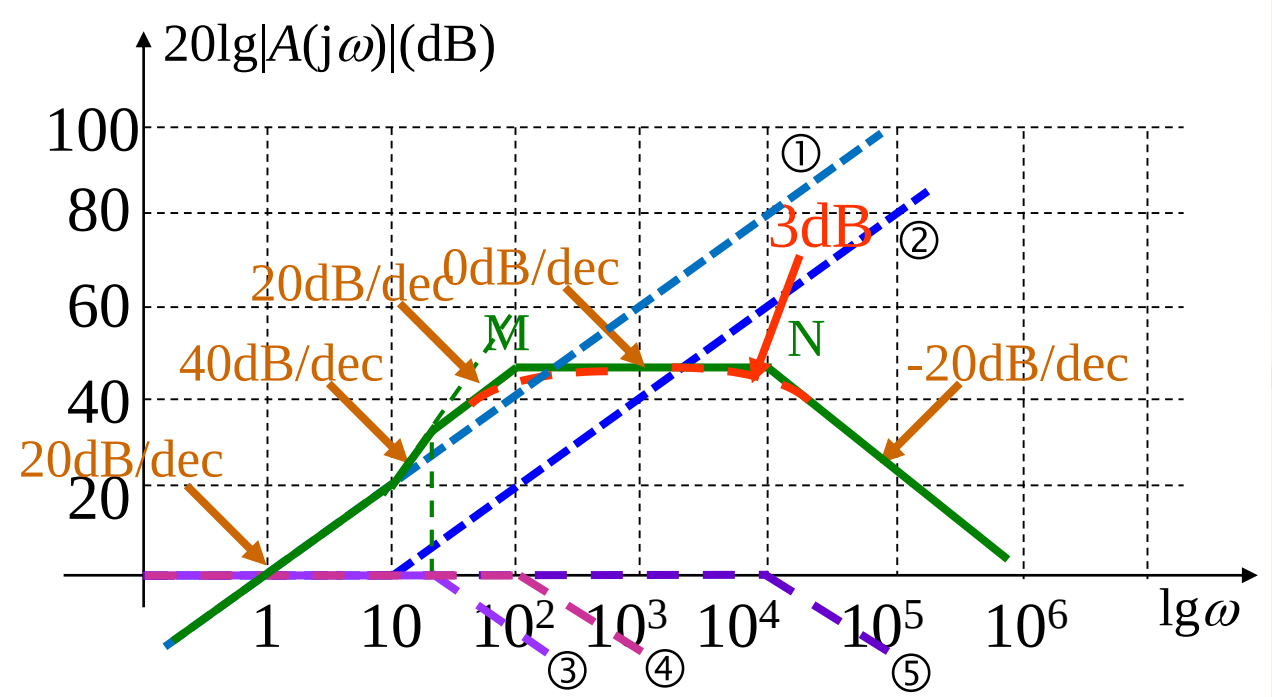
$\omega_L=10^2$

f_H -- smallest HF pole

$\omega_H=10^4$

$BW_{3dB}=\omega_H-\omega_L$

$\approx \omega_H$



5-4-2 High Frequency Model of transistors

- Mixed π Equivalent Model
- Unilateral Approximate Model
- HF Cutoff

Mixed π Equivalent Model

HF model of transistors

r_e, r_c -- bulk resistors in emitter and collector region.

$r_e < r_c$, $r_c < 10\ \Omega$. small, negligible (short)

$r_{bb'}$ -- bulk resistor in base region. $10 < r_{bb'} < 10\ \Omega$

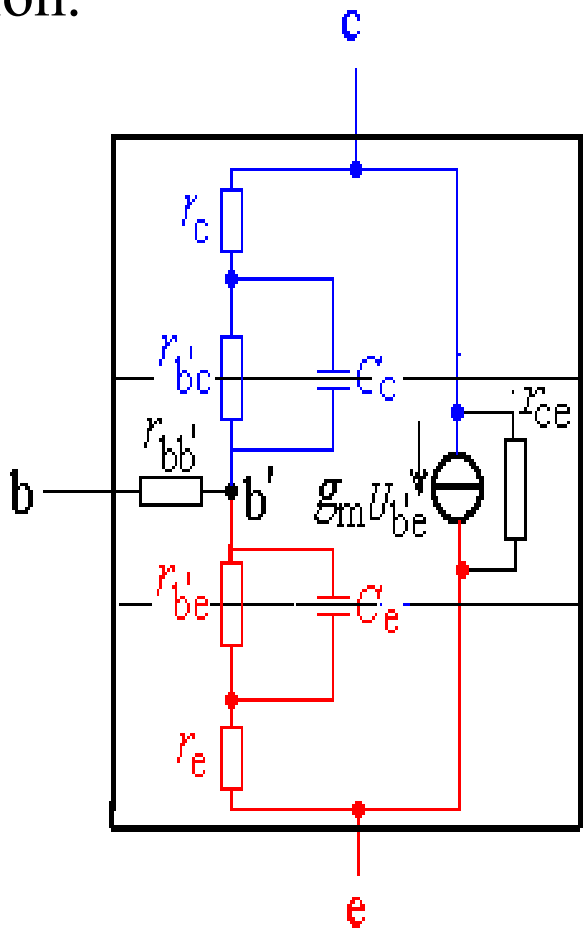
b' -- virtual equivalent point in base region

$r_{b'e}$, C_e -- equivalent AC resistor and junction capacitor of be junction

$r_{b'c}$, C_c -- equivalent AC resistor and junction capacitor of bc junction

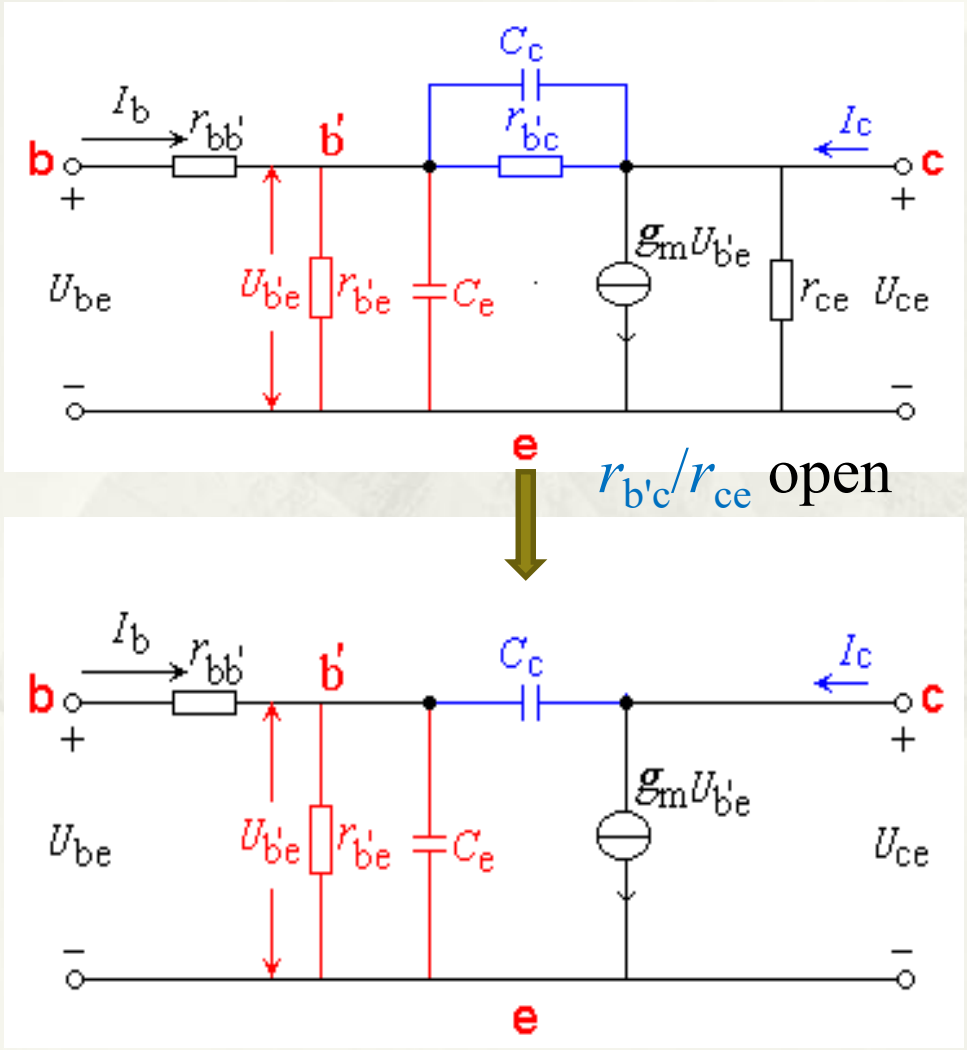
$g_m U_{b'e}$ -- dependent source

r_{ce} -- internal resistor of the dependent current source. large, negligible (open)



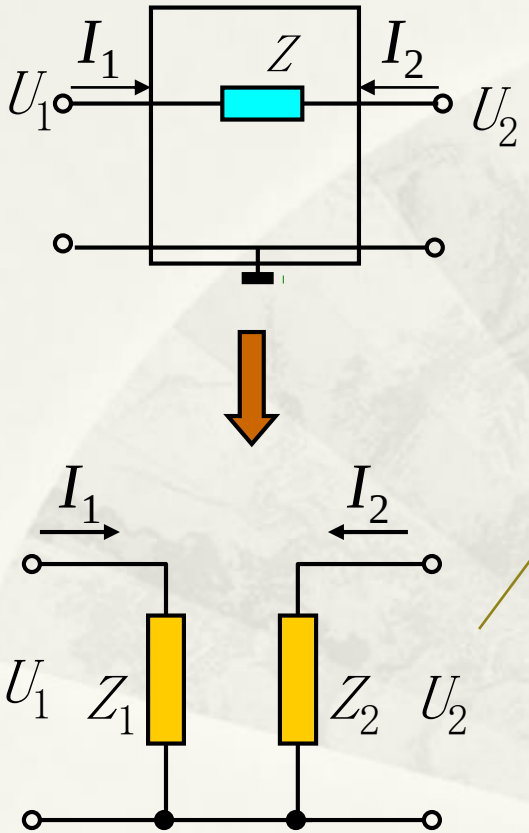
Mixed π Equivalent Model

- **HF model of transistors**



Unilateral Approximate Model

Miller Theory



Assume the gain is $K = U_2/U_1$, determine Z_1/ Z_1

$$I_1 = \frac{(U_1 - U_2)}{Z} = \frac{U_1(1-\frac{U_2}{U_1})}{Z} = \frac{\frac{U_1}{Z}}{1-K}$$

$$I_1 = \frac{U_1}{Z_1} \Rightarrow Z_1 = \frac{Z}{1-K} \qquad Z_2 = \frac{K}{K-1}Z$$

- If Z is a capacitor, $Z = 1/\omega C$:

$$C_1 = (1-K)C, \quad C_2 = \frac{K-1}{K}C$$

- If $K >> 1$, then $C_2 \approx C$

Unilateral Approximate Model

Approximate Model

$$K = \frac{U_o}{U_{b'e}} = -g_m R_L'$$

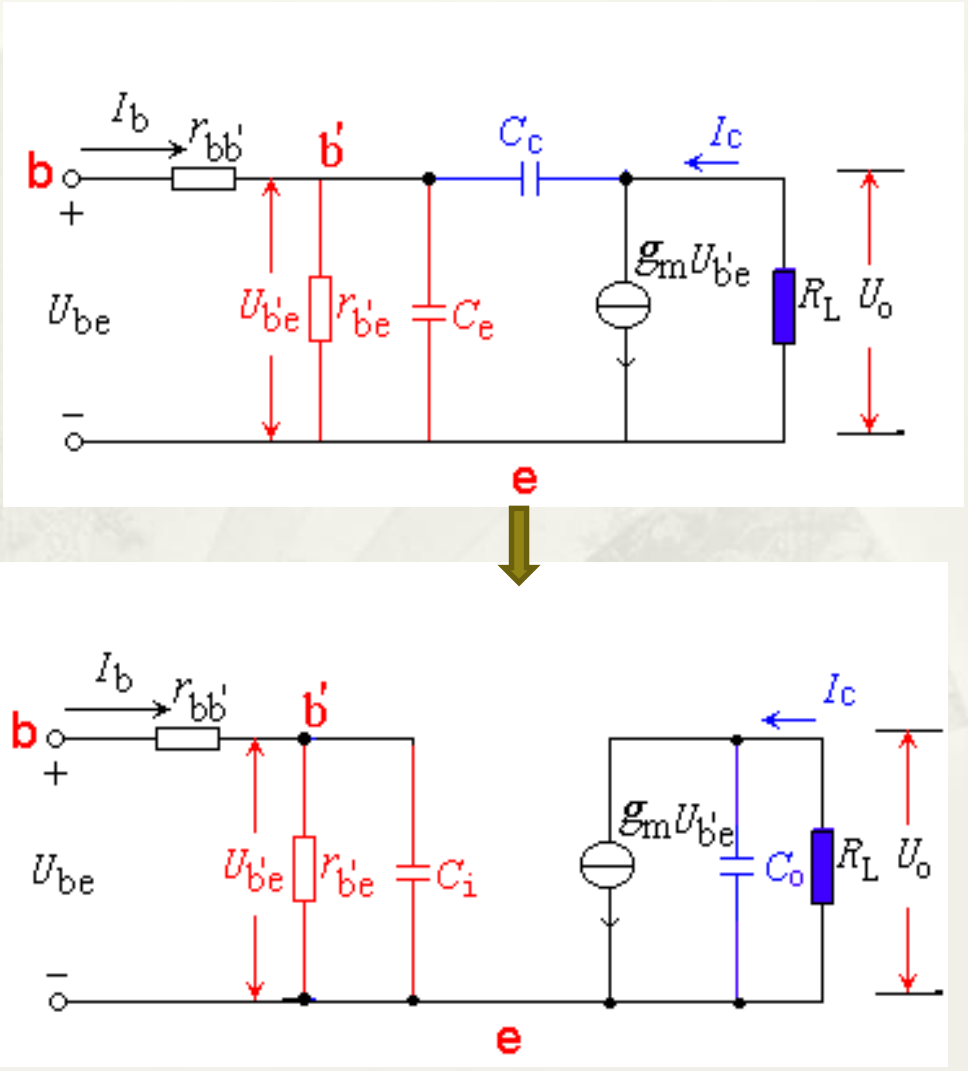
$$C_i = C_e + C_1$$

$$= C_e + (1 + g_m R_L')C_C$$

$$C_o = C_C \cdot \frac{-g_m R_L' - 1}{-g_m R_L'}$$

$$= \frac{g_m R_L' + 1}{g_m R_L'} \cdot C_C$$

$$\approx C_C$$



HF Cutoff

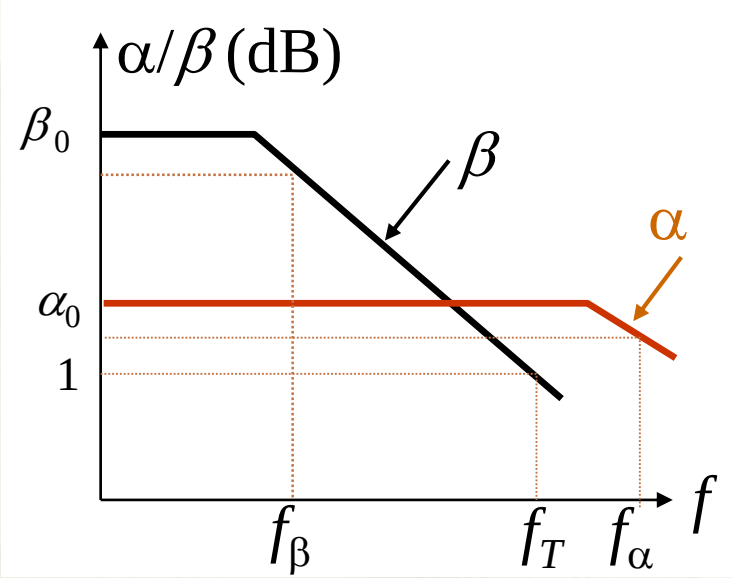
Different HF cutoff in different configuration

f_β -- CE cutoff, where $\beta = \dfrac{\beta_0}{\sqrt{2}}$

f_T -- characteristic frequency, where

$$|\beta| = 1$$

f_α -- CB cutoff, where $\alpha = \dfrac{\alpha_0}{\sqrt{2}}$



β_0 -- CE current gain in IF

α_0 -- CB current gain in IF

$f_\beta << f_T < f_\alpha$

$f_T = \beta_0 f_\beta$

HF Cutoff

- f_β

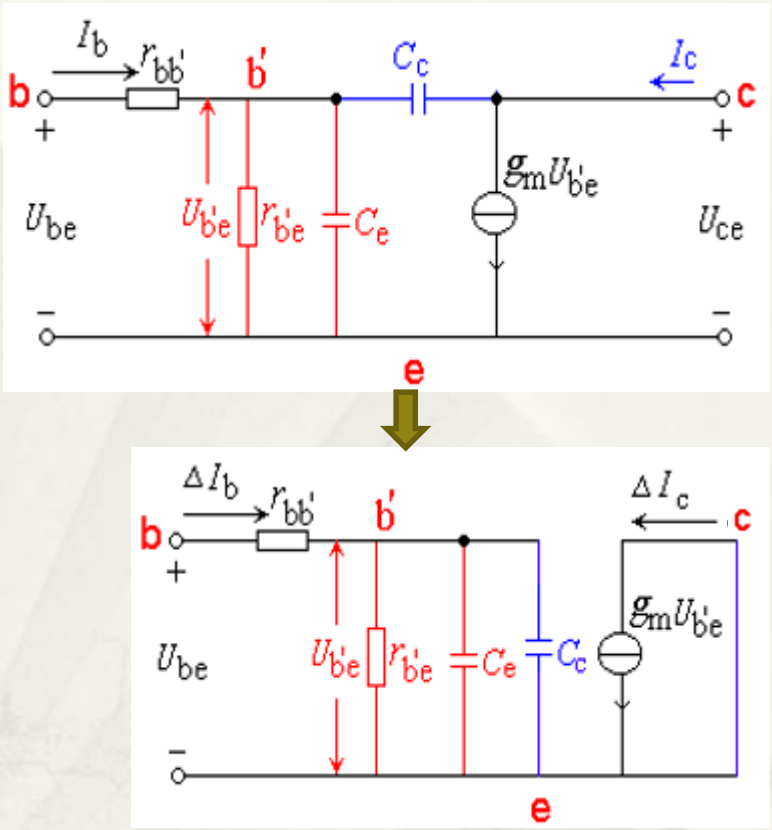
$$\beta = \frac{\Delta I_c}{\Delta I_b}\Big|\Delta U_{ce} = 0$$

$$\Delta I_c = g_m U_{b'e}$$

$$\Delta I_b = U_{b'e}[(1/r_{b'e}) + j\omega(C_c + C_e)]$$

$$\beta = \frac{g_m r_{b'e}}{1 + j\omega r_{b'e}(C_e + C_c)} = \frac{\beta_0}{1 + j\dfrac{f}{f_\beta}}$$

$$f_\beta = \frac{1}{2\pi r_{b'e}(C_e + C_c)}$$

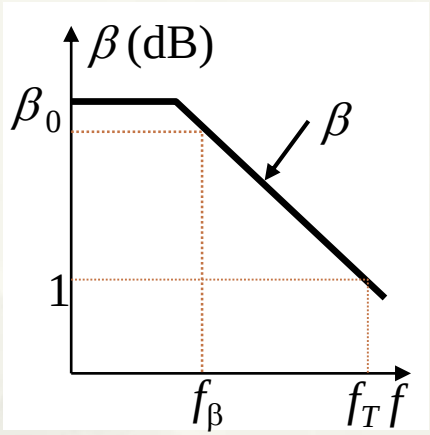


When $f = f_\beta$, $\beta = \dfrac{\beta_0}{\sqrt{2}}$

HF Cutoff

- f_T

$$\beta = \frac{\beta_0}{1 + \mathrm{j}\dfrac{f}{f_\beta}} \Rightarrow |\beta| = \frac{\beta_0}{\left|1 + \mathrm{j}f/f_\beta\right|} = \beta_0 \frac{f_\beta}{\sqrt{f_\beta^2 + f^2}}$$



When $f >> f_\beta$ $\Rightarrow$ $|\beta| \approx \beta_0 \dfrac{f_\beta}{f}$ $\Rightarrow$ $|\beta|\, f \approx \beta_0 f_\beta$

gain-bandwidth product

When $f = f_T$, $|\beta|=1$ $\Rightarrow$ $\boxed{f_T \approx \beta_0 f_\beta}$

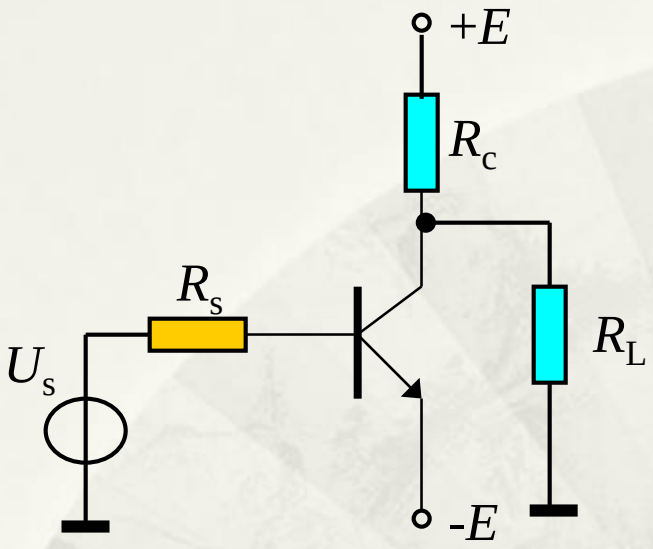
1. When $f >> f_\beta$, $|\beta| f$ keeps a constant .
   ---- $|\beta|$ will be doubled down if the operating frequency is doubled
2. f_T ---- unity-gain bandwidth
3. f_T is easy measured relatively.

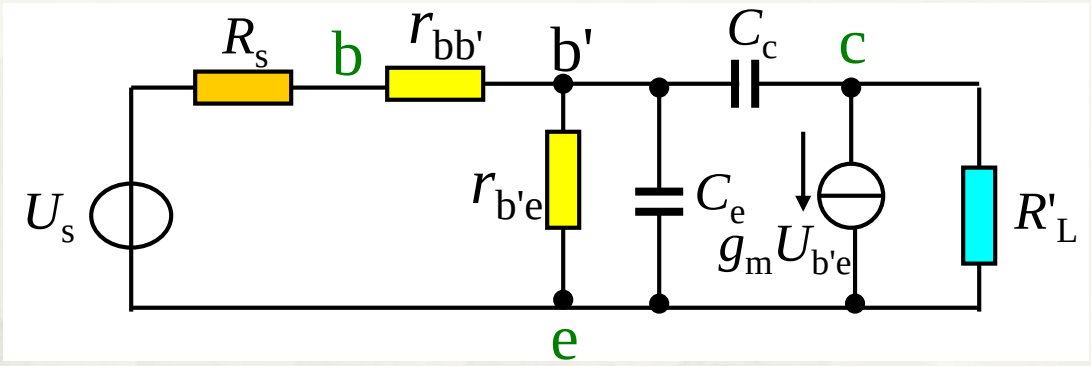
5-4-3 Frequency Response Analysis of Amplifiers

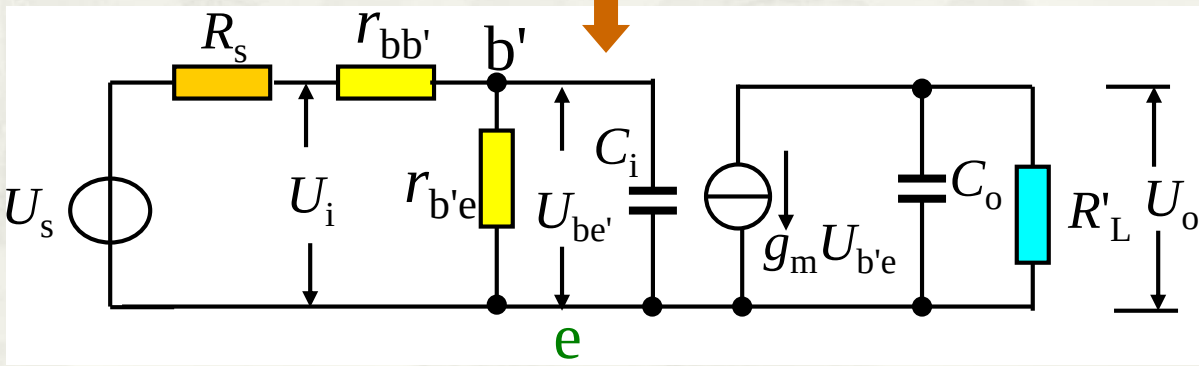
- HF Response of CE Amplifiers
- HF Response of CS Amplifiers
- Discussion

HF Response of CE Amplifiers

- **Step 1: determine the HF AC equivalent circuit**







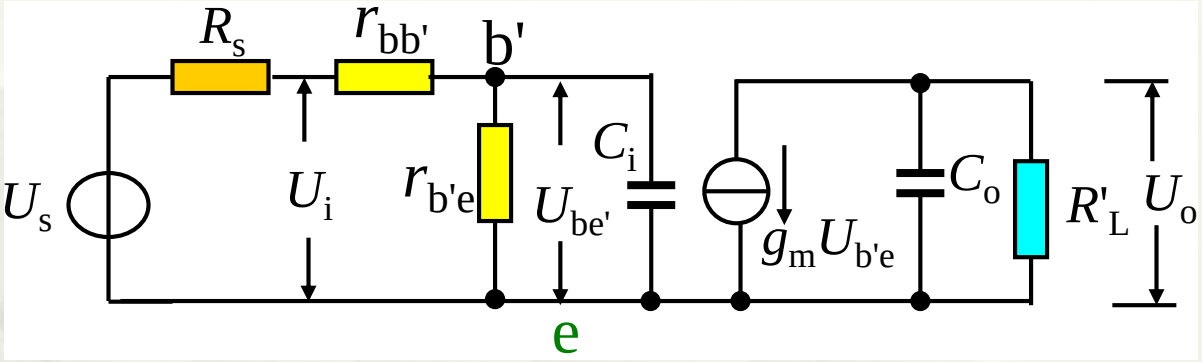
$$C_o = \frac{1 + g_m R_L'}{g_m R_L'} C_C$$

$$C_i = C_e + C_C(1 + g_m R_L')$$

$$R_L' = R_c // R_L$$

HF Response of CE Amplifiers

- **Step 2: Determine IF voltage gain**

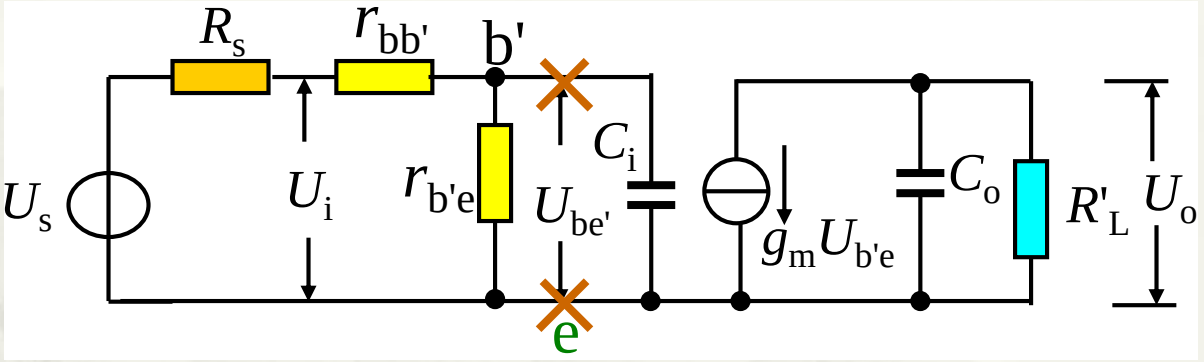


$$A_{Us} = \frac{U_o}{U_s} = \frac{U_o}{U_i} \cdot \frac{U_i}{U_s} = \frac{h_{ie}}{R_s + h_{ie}} \cdot \frac{-h_{fe} R_L'}{h_{ie}}$$

$$A_{Us} = \frac{-h_{fe} R_L'}{R_s + h_{ie}}$$

HF Response of CE Amplifiers

- **Step 3: Determine zeros and poles**

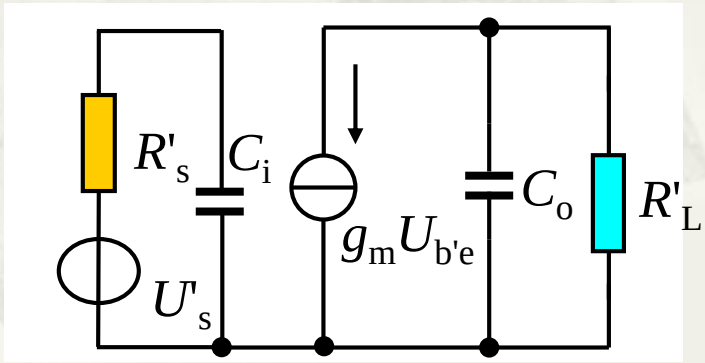


$$U_s' = \frac{r_{b'e}}{R_s + r_{bb'} + r_{b'e}} U_s$$

$$R_s' = (R_s + r_{bb'}) // r_{b'e}$$

$$\omega_{h1} = \frac{1}{R_s' C_i} \qquad \omega_{h2} = \frac{1}{R_L' C_o}$$

further simplification



HF Response of CE Amplifiers

- **Step 4: Determine HF voltage gain**

$$A_{Us}(j\omega) = \frac{A_{Us}}{(1+j\frac{\omega}{\omega_{h1}})(1+j\frac{\omega}{\omega_{h2}})} \quad \text{where} \quad A_{Us} = \frac{-h_{fe}R_L'}{R_s + h_{ie}}$$

- **Step 5: Determine f_L, f_H and *BW***

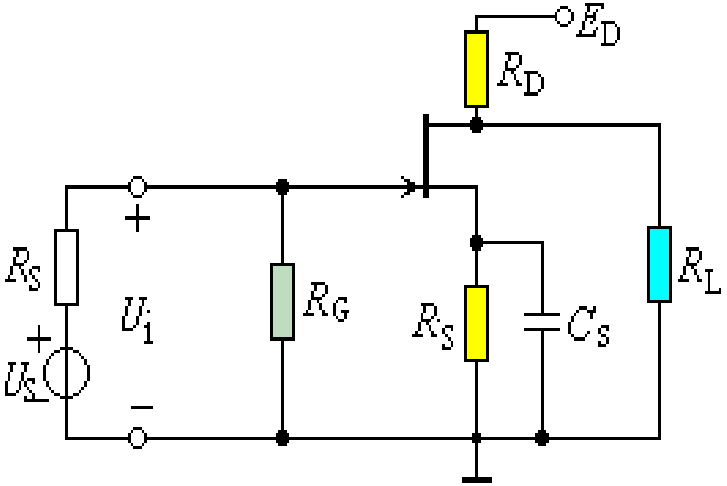
$$f_h \approx \min(f_{h1}, f_{h2})$$

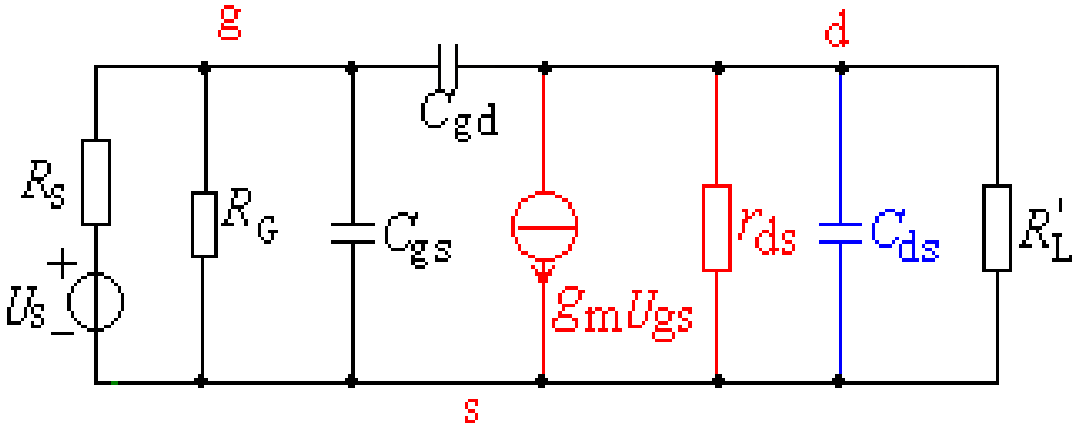
Also $\omega_h = \dfrac{1}{\sqrt{1/\omega_{h1}^2 + 1/\omega_{h2}^2}}$ generalize $\omega_h = \dfrac{1}{\sqrt{1/\omega_{h1}^2 + 1/\omega_{h2}^2 + \cdots + 1/\omega_{hn}^2}}$

$$BW = f_h - f_l \approx f_h$$

HF Response of CS Amplifiers

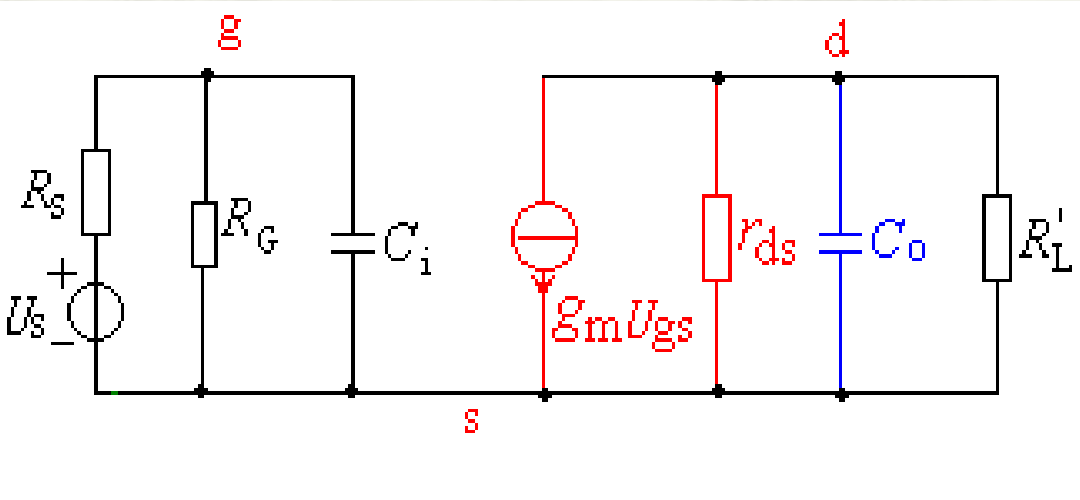
- **Step 1: determine the HF AC equivalent circuit**





$$C_i = C_{gs} + C_{gd}(1 + g_m R_L')$$

$$C_o = C_{ds} + \frac{1 + g_m R_L'}{g_m R_L'} C_{gd}$$



HF Response of CS Amplifiers

- **Step 2: Determine IF voltage gain**

$$A_{\mathrm{Us}} = \frac{R_{\mathrm{G}}}{R_{\mathrm{s}} + R_{\mathrm{G}}}(-g_{\mathrm{m}} R_{\mathrm{L}}') \qquad \text{where} \quad R_{\mathrm{L}}' = R_{\mathrm{D}} // R_{\mathrm{L}}$$

- **Step 3: Determine zeros and poles**

$$\omega_{\mathrm{h1}} = \frac{1}{(R_{\mathrm{s}} // R_{\mathrm{G}})C_{\mathrm{i}}} \qquad \omega_{\mathrm{h2}} = \frac{1}{(r_{\mathrm{ds}} // R_{\mathrm{D}} // R_{\mathrm{L}})C_{\mathrm{o}}}$$

- **Step 4: Determine HF voltage gain**

$$A_{\mathrm{Us}}(\mathrm{j}\omega) = \frac{A_{\mathrm{Us}}}{(1 + \mathrm{j}\frac{\omega}{\omega_{\mathrm{h1}}})(1 + \mathrm{j}\frac{\omega}{\omega_{\mathrm{h2}}})}$$

Discussion

- **f_H only**

---- there is only poles

$$A(\mathrm{j}\omega) = \frac{A}{(1+\mathrm{j}\frac{\omega}{\omega_{\mathrm{h1}}})(1+\mathrm{j}\frac{\omega}{\omega_{\mathrm{h2}}})\cdots(1+\mathrm{j}\frac{\omega}{\omega_{\mathrm{hn}}})}$$

$$f_{\mathrm{h}} \approx \min(f_{\mathrm{h1}}, f_{\mathrm{h2}}, \ldots, f_{\mathrm{hn}})$$

or

$$\omega_{\mathrm{h}} = \frac{1}{\sqrt{1/\omega_{\mathrm{h1}}{}^2 + 1/\omega_{\mathrm{h2}}{}^2 + \cdots + 1/\omega_{\mathrm{hn}}{}^2}}$$

Discussion

- **f_L only**

---- the number of zeros and poles are equal

$$A(\mathrm{j}\omega) = \frac{\mathrm{A}(\mathrm{j}\omega)^{\mathrm{n}}}{(\mathrm{j}\omega - p_1)(\mathrm{j}\omega - p_2)\cdots(\mathrm{j}\omega - p_{\mathrm{n}})}$$

$f_{\mathrm{L}} \approx \max(f_{\mathrm{l1}}, f_{\mathrm{l2}}, \ldots, f_{\mathrm{ln}})$

or

$$\omega_{\mathrm{l}} = \sqrt{p_1^2 + p_2^2 + \cdots p_{\mathrm{n}}^2}$$

Discussion

- f_L & f_H

---- the number of zeros m is less than the number of poles n, $m < n$

$$A(\mathrm{j}\omega) = \frac{A'(\mathrm{j}\omega - z_1)(\mathrm{j}\omega - z_2)\cdots(\mathrm{j}\omega - z_\mathrm{m})}{(\mathrm{j}\omega - p_1)(\mathrm{j}\omega - p_2)\cdots(\mathrm{j}\omega - p_\mathrm{n})}$$

Example:

$$A(\mathrm{j}\omega) = \frac{10^5(\mathrm{j}\omega)(\mathrm{j}\omega + 10)}{(\mathrm{j}\omega + 20)(\mathrm{j}\omega + 100)(\mathrm{j}\omega + 10^3)}$$

LF cutoff

$$f_\mathrm{l} = 100/2\pi$$

HF cutoff

$$f_\mathrm{h} = 10^3/2\pi$$

Discussion

Summary

- The coupling capacitors are the key reasons of causing low frequency response
- The junction capacitors and distribution capacitors of transistors are the key reasons of causing high frequency response
- gain-bandwidth product is an important parameter
- f_H of CB configuration is higher than those of CE configuration due to the small input capacitance.

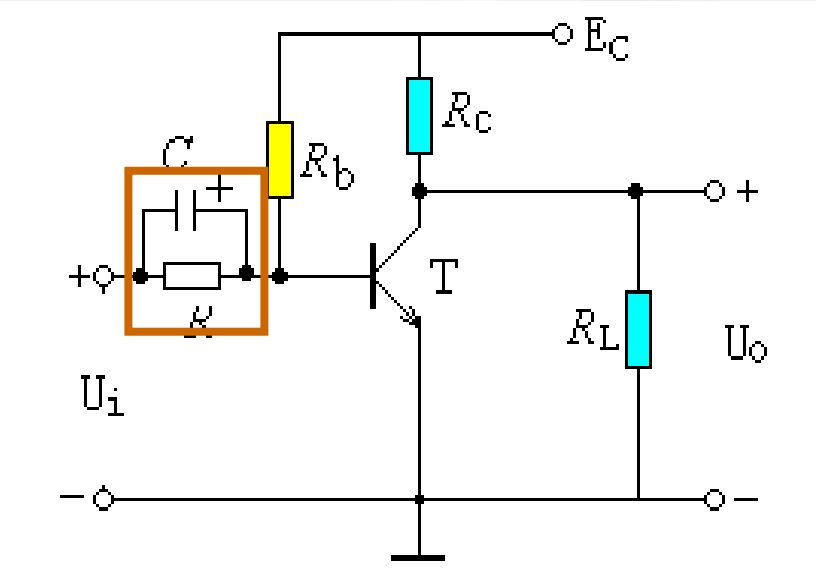
5-4-4 Methods of Expanding Bandwidth

- Compensation Circuits
- Negative Feedback
- Combinational Circuit

Compensation Circuits

Theory

---- increase HF cutoff



$$A_U = \frac{-h_{fe} R_C}{Z + h_{ie}}, \quad Z = R // \frac{1}{j\omega C}$$

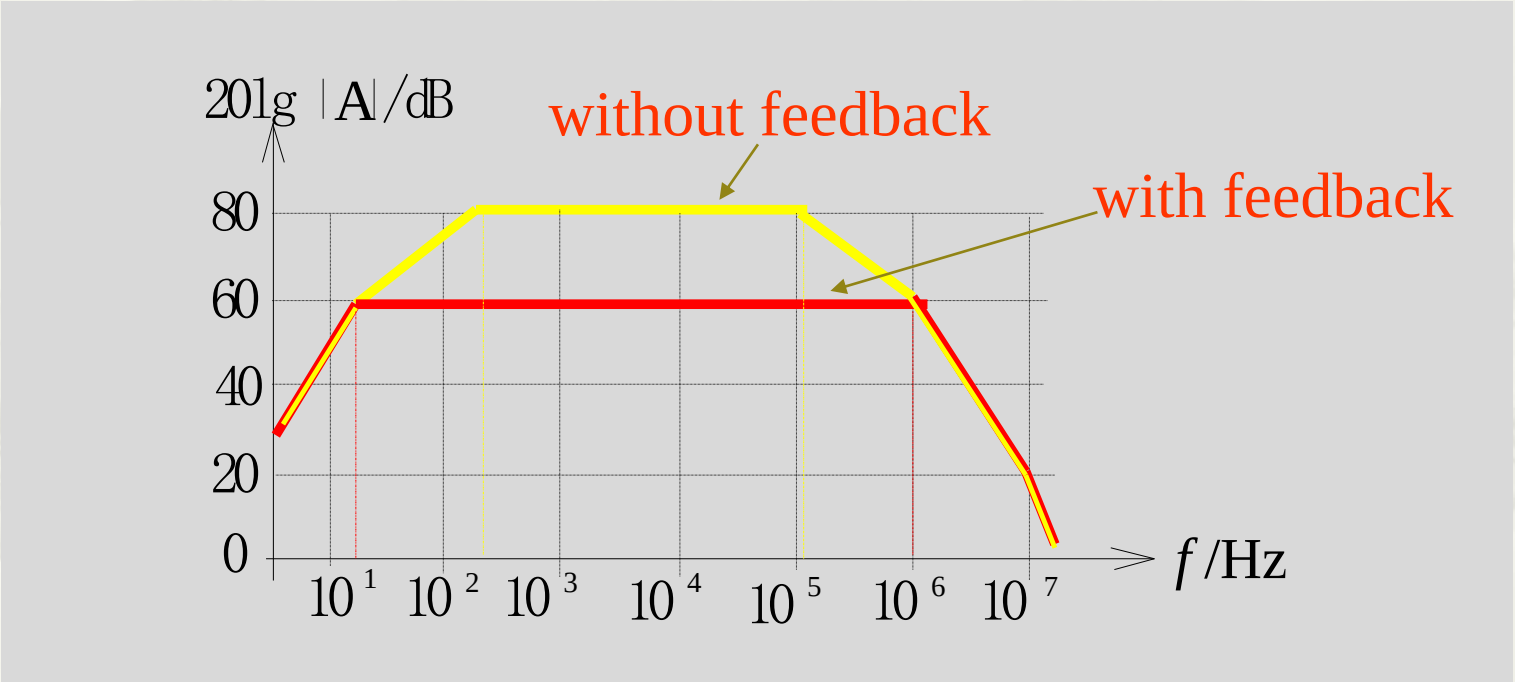
In HF, $Z\downarrow$, $A_U\uparrow$, bandwidth is widen

- ✓ Available in local widening
- ✓ Be applied when amplifiers are debugged

Negative Feedback

- **Theory**

---- voltage gain is reduced, and bandwidth is widen when negative feedback is added into amplifiers



Combinational Circuit

- **Change the time constant to increase HF cutoff.**

Example: a **CE** amplifier

$$A_{Us}(j\omega)=\frac{A_{Us}}{(1+j\frac{\omega}{\omega_{h1}})(1+j\frac{\omega}{\omega_{h2}})}$$

$$C_i=C_e+C_C(1+g_mR_L') \qquad R_S'=r_{b'e}//(r_{bb'}+R_S)$$

$$C_o=\frac{1+g_mR_L'}{g_mR_L'}C_C \qquad R_L'=R_C//R_L$$

$$\omega_{h1}=\frac{1}{R_S'C_i} \qquad \omega_{h2}=\frac{1}{R_L'C_o}$$

analysis:

$R_S\downarrow \rightarrow R_S'\downarrow \rightarrow \omega_{h1}\uparrow$ ---- add a **CC** amplifier with low output resistance, change the time constant to increase ω_{h1}

$R_L\downarrow \rightarrow R_L'\downarrow \rightarrow C_i\downarrow \rightarrow \omega_{h1}\uparrow$

$\omega_{h2}\uparrow$ ---- add a **CB** amplifier with low input resistance, change the time constant to increase ω_{h1}/ω_{h2}

5-4-5 Brief Summary

Concepts of Frequency Response

- The gain and phase of a amplifier dependents on frequency, which are named magnitude characteristic and phase characteristic respectively; Both are collectively called frequency response.
  - ✓HF response — caused by junction capacitors of transistors
  - ✓LF response — caused by coupling and bypass capacitors in circuits

Analysis of HF Response

- apply HF model -- mixed-π model
- Steps
  - ✓draw the AC path of the amplifier, then draw the HF equivalent circuit, replacing transistors to mixed-π models
  - ✓determine the IF voltage gain
  - ✓determine the HF poles
  - ✓write the HF transfer function to determine HF cutoff and bandwidth

5-4-5 Brief Summary

Widen Bandwidth of A amplifier

- Compensation circuit
- Negative feedback
- Combinational circuit

Key and Difficult points

- Key points -- analysis of frequency response
- Difficult points -- draw bode plot